Supplementary Material for *Variable social organisation is ubiquitous in Artiodactyla and likely evolved from pair-living ancestors*

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Supplementary text

S1: Explanation of dataset

The Excel spreadsheet with our dataset (https://github.com/adrianjaeggi/artiodactyl.socialorg) consists of three worksheets.

1) *Population data for analysis* - includes all populations with data on social organisation. Each row represents a population, indicated in Column D. Columns I-N and Q represent social organisations reported where numbers in each cell indicating the number of times a social organisation was reported in the population. Information in Column F was only useful determining variable social organisation in populations in which (i) solitary individuals of both sexes, (ii) pair-living, and/ or (iii) sex-specific groups were also reported. In these species, we could infer that groups of mixed sex and at least one additional form of social organization was observed. Information in Columns G and H was used to determine solitary (Column I). Information in Columns O and P was used to determine uni-sex groups (Column N). Population-level descriptions of social organisation for models 1 and 2 are shown in columns R and S, respectively. Research effort is shown in Column T.

Data that were only available at the species level (Columns U-Y) were used in Model 1 (population-level). These data are reported in worksheet 2 (*Species level summary*) because the population level dataset has rows for each species. Abbreviated references listed in Column AD are listed in Supplemental Material S5.

2) *Species level summary* – summarizes information from the *population data for analysis* worksheet. Information includes the number times social organisations were reported for each species (columns I-N, Q), number of studies per species (column T), and number of populations studied (column U). This worksheet also includes predictor variables determined at the species level (columns V-AA) as well as the types habitats (column AB) and number of habitats (column AC) summarized for the species.

3) *Code for data entry* – defines column headings and provides reference information for breeding, level of dimorphism, and body mass data.

S2. Additional details about the statistical methods

(i) Multinomial models

Similar to a logistic regression, or binomial model, where the probability p of one of two possible outcomes (e.g. a success) is estimated and the probability q of the other outcome (e.g. a failure) is then given by $q=1-p$, multinomial models estimate probabilities for k-1 of k possible states, resulting in k-1 intercepts, reflecting the probability of each outcome at baseline, and k-1 coefficients for the association between each predictor and the multinomial outcome (i.e. the change in probability of each outcome with one unit change in the predictor). The global intercept in a phylogenetic regression model represents the ancestral state [1], in this case the likelihood of different social organizations at the root of the tree, or the phylogenetically-controlled average [2, 3]. The intercepts are conditional on all predictors being at their baseline levels, i.e. the reference category for categorical predictors, or 0 for continuous predictors. Coefficients represent changes in the probability of different social organizations in seasonal vs non-seasonal breeders, when changing sexual dimorphism or female body size by one standard deviation, or when adding one habitat type or one study.

(ii) Ancestral state estimation for the predictor variables

Ancestral state estimation of species-level predictor variables was conducted using the *ape* package v. 5.2. [4] in the R statistical environment v. 3.5.1. [5]. *Ape* uses maximum likelihood estimation for discrete traits and restricted maximum likelihood for continuous traits. For discrete traits (i.e. breeding seasonality yes/no), different models of the transition between states (i.e. is no seasonality as likely to transition to seasonality as vice versa or not) were compared using likelihood ratio tests, and results from the preferred model used. For continuous traits (e.g. body size) we also compared *ape* to the fastAnc algorithm in the *phytools* package v. 0.6-44 [6]; there were no qualitative differences. Table S3 reports the summaries of these estimates, for full details please see the accompanying R code. Note that these estimated ancestral states can also be viewed as phylogenetically-controlled averages, i.e. the typical values for the extant artiodactyla [2-3].

Supplemental results

Table S1: Species level social state, indicating the extent of IVSO in Artiodactyla.

Genus Species	Species-level social state	References
Antilocapra americana	Pair-living & Group-living	[7-9]
Aepyceros melampus	Variable group-living	[10-15]
Alcelaphus buselaphus	Variable group-living	[16, 17]
Ammotragus lervia	Solitary & Group-living	[18-23]
Antidorcas marsupialis	Solitary & Pair-Living & Group-Living	[24-28]
Antilope cervicapra	Sex-specific & Group-living	[29, 30]
Beatragus hunteri	Variable group-living	[31, 32]
Bison bison	Variable group-living	[33-40]
Bison bonasus	Variable group-living	[41-44]
Bubalus bubalis	Variable group-living	[45, 46]
Bubalus depressicornis	Solitary & Pair-living	[47]
Bubalus mindorensis	Solitary & Pair-Living & Group-Living	[48-50]
Budorcas taxicolor	Solitary & Pair-Living & Group-Living	[51]
Capra ibex	Solitary & Group-living	[52-58]
Capra nubiana	Solitary & Pair-Living & Group-Living	[59-61]
Capra sibirica	Variable group-living	[62-67]
Capricornis crispus	Solitary & Pair-Living & Group-Living	[68-70]
Cephalophus silvicultor	Solitary	[71]
Connochaetes gnou	Non-variable group-living	[72-74]
Connochaetes taurinus	Non-variable group-living	[75]
Damaliscus lunatus	Solitary & Pair-Living & Group-Living	[76-78]
Damaliscus pygargus	Variable group-living	[79]
Dorcatragus megalotis	Solitary & Pair-Living & Group-Living	[80-82]
Eudorcas thomsonii	Solitary	[83]
Gazella bennettii	Pair-living & Group-living	[84, 85]
Gazella dorcas	Solitary & Group-living	[86, 87]
Gazella gazella	Solitary & Pair-Living & Group-Living	[88-90]
Gazella subgutturosa	Solitary & Group-living	[91-95]
Hemitragus jayakari	Solitary & Pair-living	[86]
Hemitragus jemlahicus	Sex-specific	[97]
Hippotragus equinus	Non-variable group-living	[98, 99]
Kobus ellipsiprymnus	Variable group-living	[15, 76, 100]
Kobus kob	Solitary & Group-living	[101-105]
Kobus leche	Non-variable group-living	[106]
Kobus vardonii	Solitary & Pair-Living & Group-Living	[14, 107, 108]
Litocranius walleri	Solitary & Group-living	[15, 109]
Madoqua kirkii	Pair-living	[110-112]
Naemorhedus goral	Solitary & Pair-Living & Group-Living	[113, 114]
Nanger granti	Solitary & Pair-Living & Group-Living	[15, 83, 115, 116]

Oreamnos americanus	Variable group-living	[117, 118]
Oreotragus oreotragus	Solitary & Pair-Living & Group-Living	[119-122]
Oryx beisa	Pair-living & Group-living	[123]
Oryx gazelle	Non-variable group-living	[124]
Ourebia ourebi	Solitary & Pair-Living & Group-Living	[76, 125-129]
Ovibos moschatus	Non-variable group-living	[130-132]
Ovis ammon	Variable group-living	[67, 133, 134]
Ovis canadensis	Solitary & Group-living	[135-139]
Ovis dalli	Non-variable group-living	[140]
Pantholops hodgsonii	Solitary & Group-living	[133, 141, 142]
Pelea capreolus	Solitary & Pair-Living & Group-Living	[129, 143-145]
Procapra picticaudata	Variable group-living	[146, 147]
Procapra przewalskii	Solitary & Group-living	[147-150]
Pseudois nayaur	Variable group-living	[151, 152]
Pseudois schaeferi	Variable group-living	[153]
Raphicerus melanotis	Solitary & Pair-living	[154]
Redunca arundinum	Solitary & Pair-Living & Group-Living	[155]
Redunca fulvorufula	Solitary & Pair-Living & Sex-specific & Group-living	[129, 156-158]
Redunca redunca	Solitary & Pair-living	[159, 160]
Rupicapra pyrenaica	Variable group-living	[161-163]
Rupicapra rupicapra	Solitary & Sex-specific	[164, 165]
Saiga tatarica	Variable group-living	[166]
Sylvicapra grimmia	Solitary & Pair-living	[167, 168]
Syncerus caffer	Solitary & Group-living	[169-172]
Tetracerus quadricornis	Solitary & Pair-living	[173]
Tragelaphus buxtoni	Solitary & Group-living	[174, 175]
Tragelaphus eurycerus	Solitary & Group-living	[176, 177]
Tragelaphus imberbis	Non-variable group-living	[15]
Tragelaphus scriptus	Solitary & Pair-Living & Group-Living	[178-181]
Tragelaphus spekii	Variable group-living	[182]
Tragelaphus strepsiceros	Solitary & Group-living	[183-185]
Camelus bactrianus	Variable group-living	[186]
Lama guanicoe	Variable group-living	[187-189]
Vicugna vicugna	Non-variable group-living	[190-194]
Alces alces	Solitary & Group-living	[195-199]
Axis axis	Variable group-living	[200, 201]
Axis porcinus	Solitary & Pair-Living & Group-Living	[202, 203]
Blastocerus dichotomus	Solitary & Pair-Living & Group-Living	[204, 205]
Capreolus capreolus	Solitary & Group-living	[206-208]
Capreolus pygargus	Solitary & Group-living	[209]
Cervus elaphus	Solitary & Group-living	[210-215]
Cervus nippon	Solitary & Group-living	[216-218]
Dama dama	Solitary & Group-living	[219-223]
Hippocamelus antisensis	Solitary & Group-living	[224, 225]

Hippocamelus bisulcus	Solitary & Pair-Living & Group-Living	[226-228]
Muntiacus muntjac	Solitary & Pair-living	[202, 229-231]
Odocoileus hemionus	Pair-living & Group-living	[232-235]
Odocoileus virginianus	Solitary & Group-living	[233, 236-242]
Ozotoceros bezoarticus	Solitary & Group-living	[243-245]
Rangifer tarandus	Variable group-living	[246-248]
Rucervus duvaucelii	Non-variable group-living	[249]
Rusa unicolor	Variable group-living	[202]
Giraffa camelopardalis	Solitary & Group-living	[15, 250-258]
Hippopotamus amphibius	Sex-specific & Group-living	[259, 260]
Moschus cupreus	Solitary	[261]
Moschus leucogaster	Solitary	[262]
Babyrousa babyrussa	Solitary & Pair-Living & Group-Living	[263, 264]
Phacochoerus aethiopicus	Solitary & Group-living	[265]
Phacochoerus africanus	Solitary	[266, 267]
Potamochoerus porcus	Solitary & Group-living	[268]
Sus scrofa	Solitary & Group-living	[269-273]

Table S2: Expanded descriptive table for artiodactyl social organization.

Values at the species level. IVSO was inferred when a species had either more than one form of social organisation in one population, or when two populations showed different forms of social organisation. Of the 83 species with variable social organisation, seven (8.4%) showed variation between populations, thirty (36.1%) showed variation within a population, and forty-six (55.4%) showed variable social organisation both between populations and within a population.

Category	***No. and percentage***
Solitary	5 (5%)
Pair-living	1 (1%)
Sex-specific	1 (1%)
Non-variable Group-living	10 (10%)
Variable Group-living[a]	22 (22%)
Solitary & Pair-living	7 (7%)
Solitary & Group-living	26 (26%)
Solitary & Sex-specific	1 (1%)
Pair-living & Group-living	4 (4%)
Sex-specific & Group-living	2 (2%)
Solitary, Pair-living & Group-living	20 (20%)
Solitary, Pair-living, Sex-specific & Group-living	1 (1%)
Total number of species showing any form of variable social organisation, including variable group composition	83 (83%)
Species with data	100
Species with no data	126

[a] *Referring to the basic sex composition of adult group members; variable includes more than one of the following: single male, multiple females (MFF); single female, multiple males (FFM); multiple males, multiple females (MMFF); unisex groups, both male-only and female-only in the same population*

Table S3: Summaries of ancestral state estimations of species-level predictors

Trait	Estimate	Lower 95% CI	Upper 95% CI
log Body mass [kg]	4.19	1.76	6.62
Sexual dimorphism [M:F mass]	1.21	0.53	1.90
Breeding seasonality	Likelihood of non-seasonal: 0.502, likelihood of seasonal: 0.498		

Table S4. Likelihood of SOs for predictors at lower 95% CI level (see Table S3)

	mean	lower 95% CI	upper 95% CI
Solitary	0.09	0.00	0.52
Pair-living	0.51	0.00	1.00
Sex-specific	0.04	0.00	0.09
Group-living	0.01	0.00	0.03
Variable	0.34	0.00	0.97

Table S5. Likelihood of SOs for predictors at upper 95% CI level (see Table S3)

	mean	lower 95% CI	upper 95% CI
Solitary	0.09	0.00	0.40
Pair-living	0.00	0.00	0.00
Sex-specific	0.03	0.00	0.03
Group-living	0.38	0.00	0.98
Variable	0.50	0.00	0.98

Figure S1: SO at ancestral state and phylogenetic mean (Model 2)

Same as Figure 2 in the main text, but with "variable SO" split into five different forms to test whether pair-living was an important component for variable species. Phylogeny of artiodactyls with data on social organisation (SO), wherein the coloured boxes at the tips of the phylogeny show SOs observed in different populations of extant species. As a reference, the nine possible states (solitary, pair-living, sex-specific, group-living, pair-group, solitary-group,solitary-pair, solitary-pair-group, variable group) are plotted above and below the phylogeny in this order. Using the same colours, the inset figures show the probabilities (from Model 2) of each SO (Sol=Solitary, Pair=Pair-living, SS=Sex-specific, Grp=Non-variable group-living, PrGp=Pair-group, SlGp=Solitary-group, SlPr=Solitary-pair, SPG=Solitary-pair-group, VaGp=Variable group) when body size and dimorphism are at the levels known from the fossil record ("ancestral state"), or the phylogenetically-controlled averages estimated from extant species ("phylogenetic mean"); number of habitats and studies as well as breeding seasonality are kept at their baselines of 1 and non-seasonal, respectively. The scale bar has been omitted here.

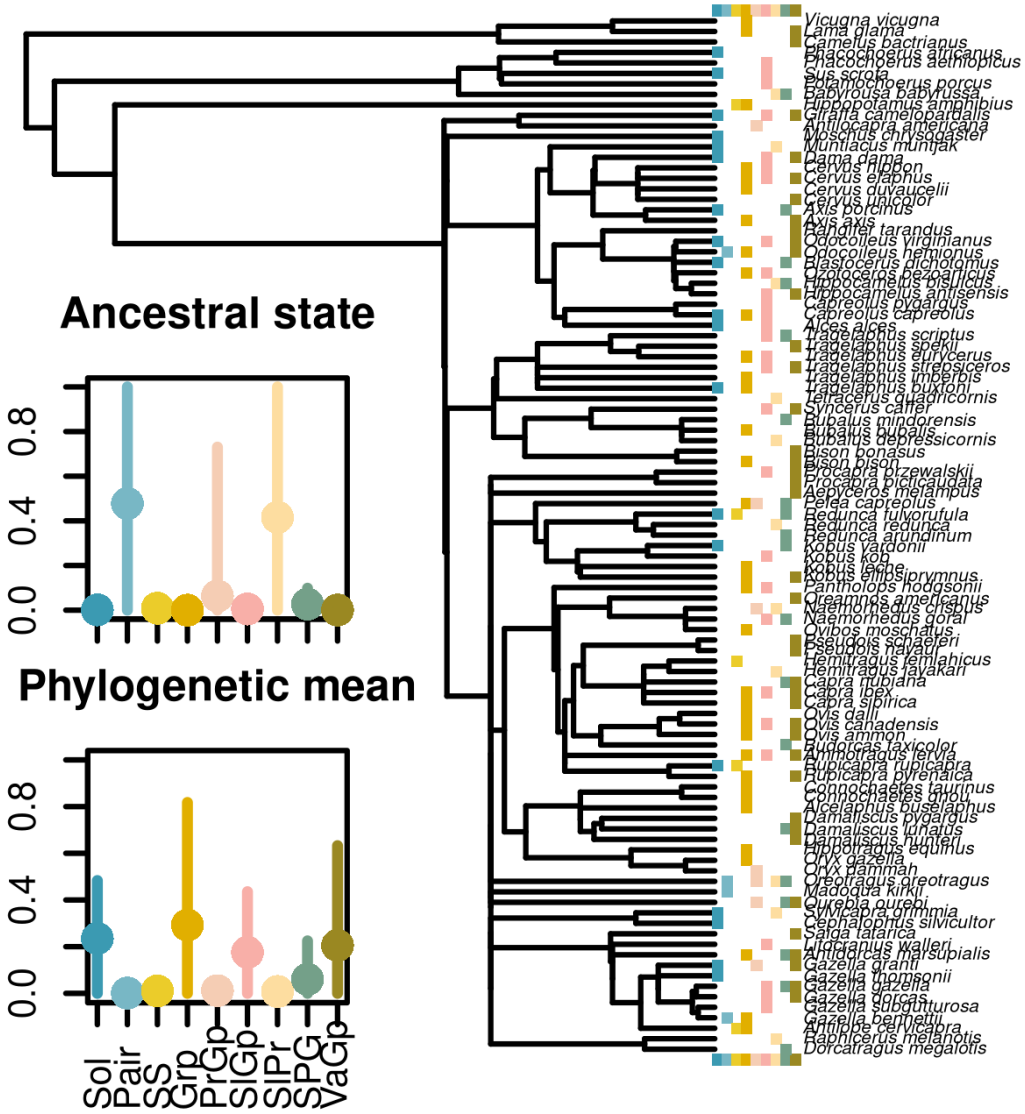
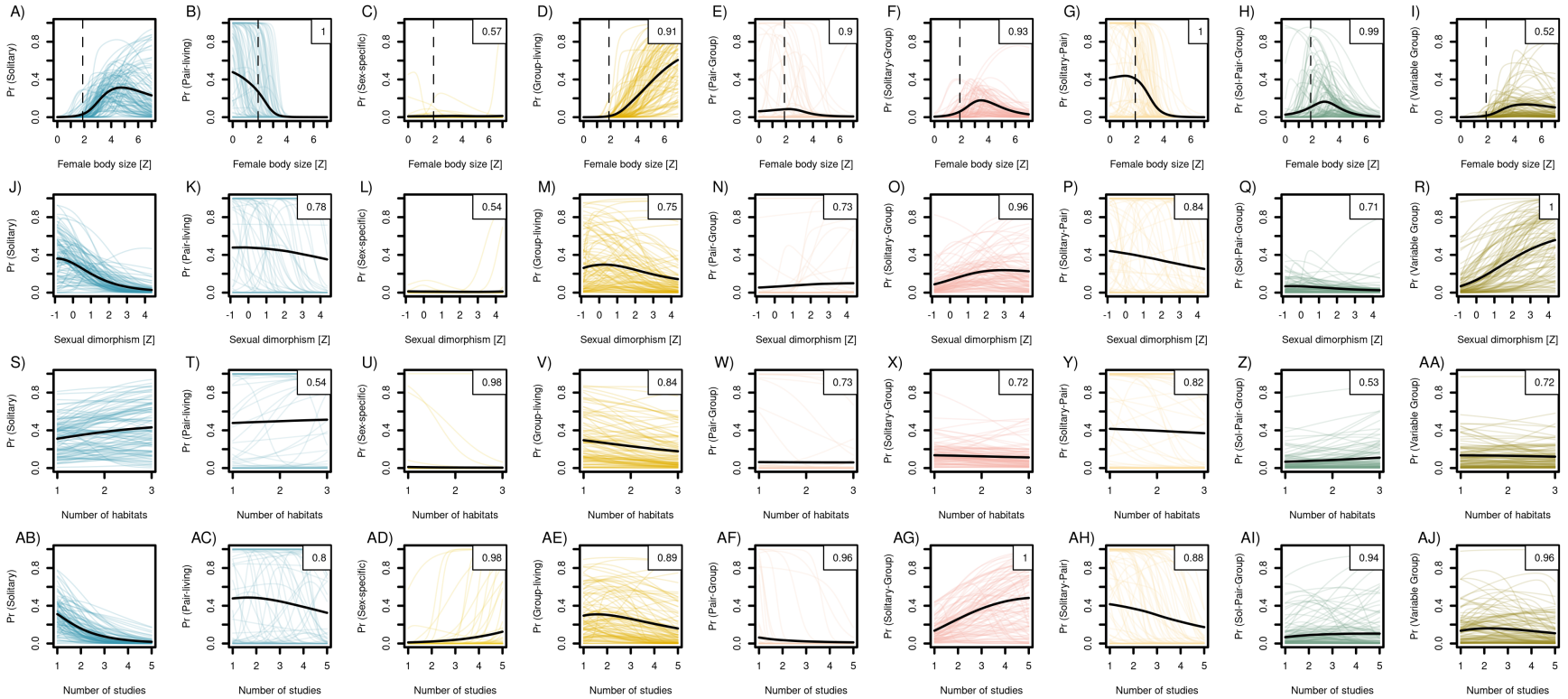


Figure S2: Probabilities of SOs as a function of predictors (Model 2)

Columns show (from left to right) the probability of solitary, pair-living, sex-specific, group-living, pair-group, solitary-group, solitary-pair, solitary-pair-group, and variable group– using the same colours as in Figure S1 – while rows show (from top to bottom) predicted changes in those probabilities as a function of female body size (A-I), sexual dimorphism (J-R), number of habitats (S-AA) and number of studies (AB-AJ). The numbers in the legends are the posterior probabilities [PP], i.e. the proportion of the posterior distribution that supports a given association; these were not available for solitary, as this was the reference category. Within each row, all other predictors were held at their baseline value, except for body size; for pair-living, pair-group, and solitary-pair body size was kept at the ancestral state, for all others it was kept at the phylogenetic mean for better visibility. Solid black lines are the predicted means, thin coloured lines are 100 random samples drawn from the posterior to illustrate uncertainty.



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